Dataset of three Cancer Diseases

STA314H1S Final Project Report

Kaggle Team: totrainornottotrain26

Ranking #6; Score: 1.00

Members:

Jiurong Han 1009842351

Jinghao Hu 1009898979

Xiaorui Huang 1009839463

Ruoxuan Fu 1009967519

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2. Problem Statement

The goal of this project is to develop a supervised classification model that best accurately predicts within three cancer types (GBM, LUSC, and OV) using high-dimensional gene data. Due to the complex nature of gene data, we have more gene features than observations. The primary challenge is identifying the most informative features while ensuring the generalizability of the model and avoiding overfitting. Motivating us to effectively perform feature selection and build a predictive model that has strong generalization performance while maintaining interpretability.

This problem is of substantial importance in both statistical and medical applications. From a medical perspective, accurate cancer classification based on a small set of features can support early diagnosis, improve treatment decisions, and enhance patient outcomes at an affordable cost. Prior studies have identified the mechanisms certain genes play in the progression and development of cancers (Au et. al., 2004; Ding et. al., 2026; Ho et. al., 2021). This evidence provides biomedical support in using gene expression data in cancer classification problems, and motivates the identification of important gene features using statistical modelling. From a statistical perspective, Lots of previous studies have shown that regularization methods such as LASSO, are effective in resolving high-dimensional genomic data by estimating a model while performing variable selection (by regularization), our project addresses key challenges in high-dimensional data analysis, such as feature selection, model generality, interpretability, and robustness in the presence of noisy or potentially mislabeled data.

3. Statistical Analyses

Data Introduction:

Name: *Dataset of three Cancer Diseases*

Source: The dataset was provided by STA314 instructor Prof. Bin

Origin: Data set derived from The Cancer Genome Atlas (TCGA), a collaborative project between the National Cancer Institute (NCI) and the National Human Genome Research Institute (NHGRI). TCGA provides large-scale genomic data that document key genetic changes across multiple cancer types and has been widely used in cancer research.

This project focused on 3 cancer types: glioblastoma multiforme (GBM), lung squamous cell carcinoma (LUSC), and ovarian cancer (OV). The dataset contains 12,043 gene expression features and 886 samples in total. In detail, the dataset includes 376 samples for GBM, 90 samples for LUSC, and 420 samples for OV. However, the data has been modified to simulate real-life data noise, the observations label has been flipped to: 408 samples for class 3, 348 samples for class 1, and 130 samples for class 2. Since the correspondence between numerical class label (i.e. class 1/2/3) and actual cancer types (i.e. GBM, LUSC, OV) is not given, one of our tasks is to identify this relationship so that our result and conclusion have real-world significance.

3.1. Exploratory Data Analysis

Our initial data exploration indicates 886 categorical responses with 3 classes, and 12044 continuous predictors. Considering the model interpretability, we are trying to match cancer types with numerical class labels by observing the distribution of important features for each class supported by external biomedical literature.

As shown in Figure 1, the distribution of TP63 exhibits a significantly higher expression level in class 2 than in other classes. Prior studies (Au et. al., 2004) have established a strong association between TP63 and LUSC. Combined with the fact that class 2 has the smallest sample size in the training data, this provides supporting evidence that class 2 corresponds to LUSC. Similarly, C1QL1 exhibits significantly higher expression in class 1. According to Ding et. al. (2026), C1QL1 was linked to GBM; we infer that class 1 most likely represents GBM. Consequently, class 3 is inferred to correspond to OV. Assuming noise is constant across classes, the inferred class ratio (GBM:LUSC:OV = 348:130:408) aligns reasonably well with the reported distribution (376:90:420), further supporting our classification.

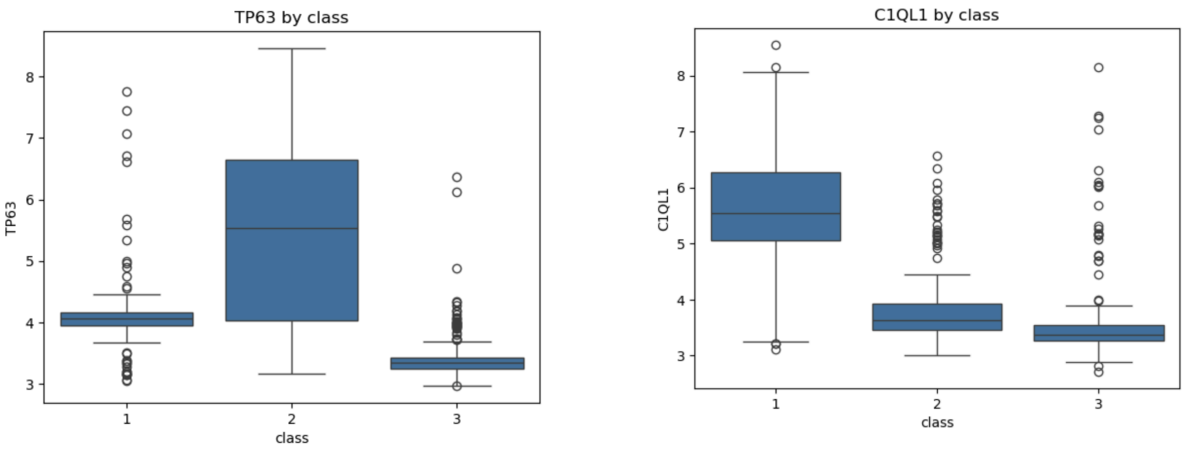

Figure 1. Gene Expression Level Distribution by Class

3.1.1. Data Preprocessing

We first examined the dataset for missing values. Checking for data validity, make sure there are no null values, ensuring data quality and avoiding potential bias or instability in model estimation. Then we prepare our data in the following steps:

- <u>Train–test split for validation:</u>
  For future evaluation of model performance, we randomly split the provided training dataset into 80% training data and 20% validation data, while keeping each subset's class proportions consistent with the original dataset by applying a stratified sampling method, ensuring a balanced representation of the response variable.
- <u>Feature standardization:</u>
  Initial exploration of the dataset showed that our predictors are small in magnitude and have relatively high variances. Due to the high-dimensionality of our data, motivating us to apply LASSO feature selection, since the LASSO method is sensitive to the scale of predictors, we standardized each feature $\frac{x-\mu}{\sigma}$ so that each predictor has a mean of 0 and a variance of 1, as shown in Figure 2. This ensures that all predictors are placed on the same scale, allowing the LASSO regularization penalty to treat each variable fairly and preventing variables with larger scales from dominating the coefficient estimates.

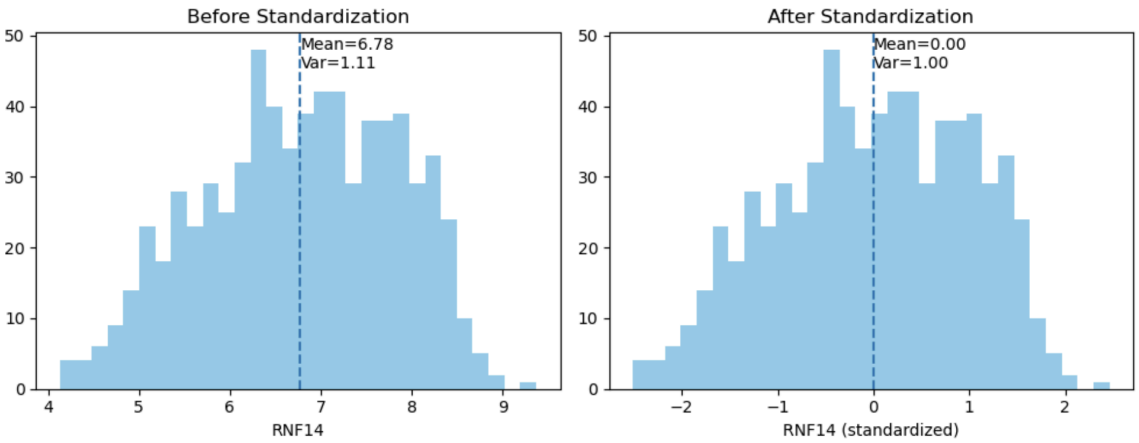

Figure 2. Example Feature RNF14 Before-After Standardization Histogram

- <u>LASSO Feature selection:</u>
  This high-dimensional setting may lead to overfitting and unstable coefficient estimates. To address this issue, we applied LASSO (Least Absolute Shrinkage and Selection Operator) for feature selection. This method preserves the most relevant features while improving model interpretability and generalization performance. Using the LASSO feature selection has been widely used in high-dimensional biomedical data analysis, especially in studies that involve gene studies, where the complexity of the gene and the number of gene features will normally far exceed the number of

observations. Prior studies have shown that LASSO methods can successfully identify vital genes in microarray data and improve predictive performance in cancer classification tasks (Ma et al., 2007). LASSO introduces a penalty term $\lambda \sum_{j=1}^{p} |\beta_j|$, on the regression coefficients, which can provide a sparse model by allowing some coefficients to shrink to 0. After LASSO feature selection we successfully reduce the number of features to 65 in total, below are the .

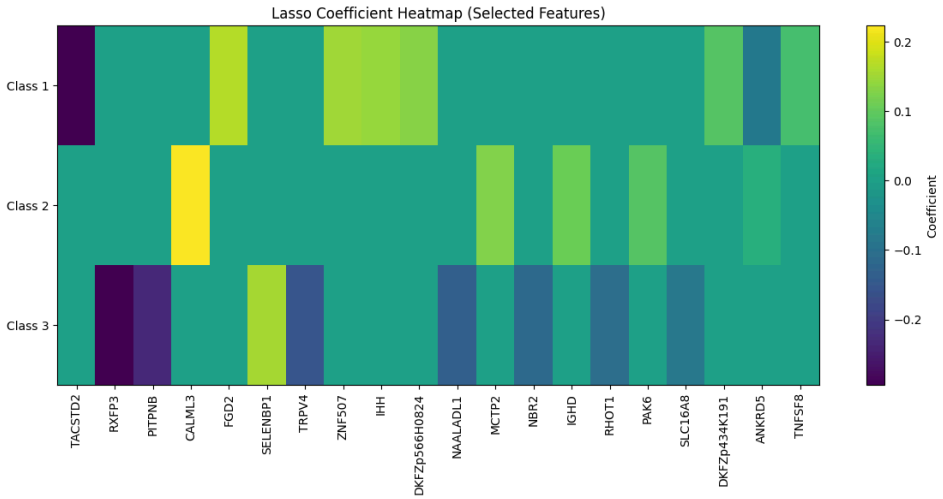

Figure 3. Top 20 Lasso-Selected Features Ranked by Magnitude Across Classes

3.2 Model Selection

We consider 4 preliminary models 1. Logistic model with Ridge regularization; 2. Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA); 3. Random Forest (RF); 4. Extreme Gradient Boosting (XGBoost). These models were considered as candidate models since linear models, generative classifiers and ensemble learning tree methods, all of which have been widely used in high-dimensional biomedical studies. We train all the candidate models on 80% training data, for those models need to find the best hyperparameter, we apply cross validation grid search to find the optimal parameter. To evaluate their performance, we perform cross-validation (CV) based on accuracy to compare models. This accuracy-based CV allowed us to find models with high predicted accuracy and generalizability. Based on the CV results, we selected the top performing model for further exploration.

3.2.1 Cross-Validation Based on Accuracy

We evaluated multiple classification models, including Logistic Regression, LDA, QDA, Random Forest, and XGBoost, using 5-fold stratified cross-validation. StratifiedKFold was used to ensure that class proportions were preserved across folds, providing a reliable estimate of model performance. By applying 5-fold CV, we obtain a robust estimate of generalization performance while preserving class proportions across folds which prevents overfitting.

Metrics such as mean squared error were not used because they are more appropriate for regression problems rather than classification tasks. In a multi-class classification setting, accuracy and confusion matrix-based evaluation provide more interpretable measures of performance.

Based on the cross-validation results, tree-based ensemble methods, particularly Random Forest and XGBoost, achieved the highest predictive performance, with accuracies close to 0.90. Random Forest achieved the highest mean accuracy, while XGBoost performed closely.

Compared to linear models such as Logistic Regression and LDA, both Random Forest and XGBoost demonstrated superior performance. Therefore, Random Forest and XGBoost were selected as the two most competitive models for further analysis. We next present the training procedure of the Random Forest model in detail.

Model	CV Mean Accuracy	CV Std
Random Forest (M8)	0.8995	0.0207
XGBoost (M7)	0.8973	0.0210
LDA (M5)	0.8961	0.0211
QDA (M6)	0.8837	0.0132
Logistic Regression (M1–M4)	0.8645	0.0203

Table 1.Mean Accuracy and Standard Deviation from 5-Fold Cross-Validation

3.2.2 Random Forest

Random forest fits multiple decision trees on bootstrap samples of the training data and combines their predictions by majority vote. At each split, it considers only a random subset of predictors instead of the full predictor set which reduces the correlation across trees and improves generalization.

Random forest is quite suitable in the cancer dataset, because it is well suited to high-dimensional multi-class classification and can flexibly capture nonlinear relationships and interactions among genes. By averaging across trees, it reduces variance and mitigates overfitting. In addition, the model provides variable importance measures, making it useful not only for prediction but also for identifying genes that contribute to distinguishing among GBM, LUSC, and OV (Alladio et al., 2024).

Random Forest Training and Hyperparameter Tuning

The Random Forest model was trained on selected training data with hyperparameter tuning via g grid search. Out-of-bag (OOB) accuracy was used as the primary selection criterion, while cross-validation used to further assess the choice of minimum leaf. The final parameter combination was selected to balance predictive performance and overfitting.

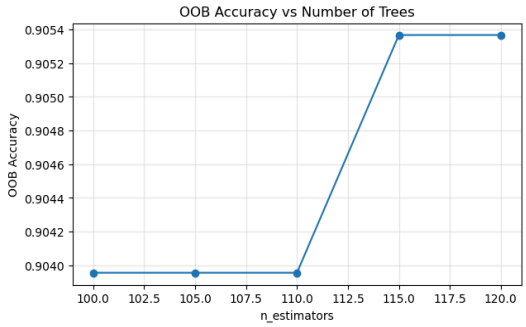

Figure 4. OOB Accuracy vs Number of Trees

	n_estimators	oob_accuracy	oob_error
0	100	0.903955	0.096045
1	105	0.903955	0.096045
2	110	0.903955	0.096045
3	115	0.905367	0.094633
4	120	0.905367	0.094633

Table2. OOB Accuracy and Error vs Number of Trees

The number of trees was selected OOB accuracy sweep over a coarse grid from 50 to 1000 trees. Since OOB observations arise naturally from bootstrap sampling, they provide an internal estimate of prediction error without requiring a separate validation set. We found that OOB accuracy increased up to around 115 trees and then plateaued. As further increases were negligible gains while computational cost continued to increase, we selected 115 trees as the optimal choice.

	max_depth	oob_accuracy	oob_error
0	1	0.809322	0.190678
1	2	0.903955	0.096045
2	3	0.903955	0.096045
3	4	0.905367	0.094633
4	5	0.905367	0.094633
5	None	0.905367	0.094633

Table 3. OOB Accuracy and Error vs Max Depth

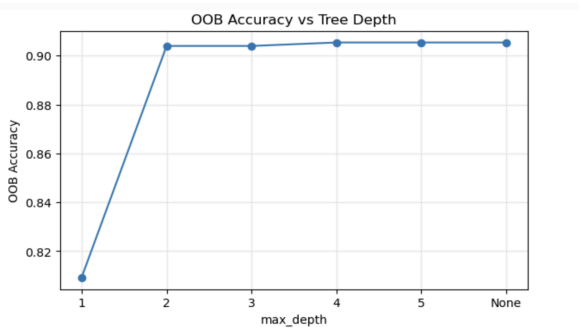

Figure 5. OOB Accuracy vs Max Depth

To choose the maximum tree depth, we performed an OOB accuracy sweep while fixing the number of trees at 115. We considered a range of candidate depths, from shallow to deeper trees (from 3 to 20). As

shown in Figure 5, a depth of 1 performed poorly because of underfitting. Increasing the depth to 2 substantially improved OOB accuracy, and the accuracy reached its peak at depth 4 before plateauing. To avoid unnecessary complexity and reduce the risk of overfitting, we selected modest depth 4.

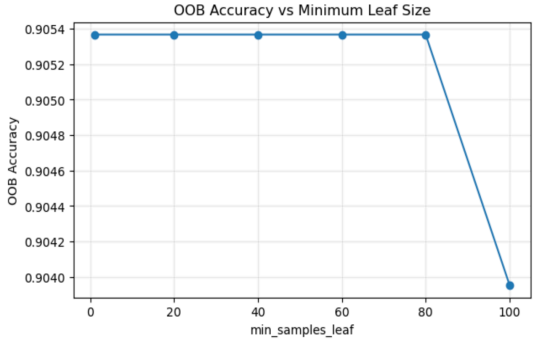

Figure 6. OOB Accuracy vs Min Leaf Size

	min_samples_leaf	oob_accuracy	oob_error
0	80	0.905367	0.094633
1	82	0.905367	0.094633
2	84	0.903955	0.096045
3	86	0.903955	0.096045
4	88	0.903955	0.096045
5	90	0.903955	0.096045

Table 4. OOB Accuracy and Error vs Min Leaf Size

We performed an OOB sweep over the range from 0 to 100 and initially identified a cutoff at minimum leaves as 82. However, a large leaf size reduces model flexibility and leads to underfitting, resulting in lower accuracy. By contrast, very small leaf sizes may increase variance and cause overfitting. Therefore, we selected minimum leaves of 42 as a trade-off between complexity and predictive performance.

	Model	CV Mean Accuracy	CV Std
0	Random Forest (M8_42)	0.899530	0.020724
1	Random Forest (M8_2)	0.899530	0.020724
2	Random Forest (M8_82)	0.898407	0.019953

Table 5. Cross-Validation Accuracy Comparison across Different Minimum Leaf Sizes

Compared to minimum leaves as 82 (shown in Table 5), cross validation results show that minimum leaves of 42 achieves higher accuracy, although with marginally higher variance. Since a minimum leaf size of 2 does not provide any additional improvement in accuracy, and may increase the risk of overfitting, we excluded such small leaf sizes from consideration. Accordingly, we selected n_estimators=115, max depth = 4, and min_samples_leaf = 42 as the final hyperparameter settings.

4. Results and Conclusion

4.1 Results

4.1.1 Evaluating the Generalization Performance of Random Forest

To further compare the performance of Random Forest and XGBoost, we examined their confusion matrices on both training and test datasets. The Figure7 shows that both models achieve similar classification performance, with most predictions concentrated along the diagonal. While accuracy is the primary metric which we used to select models from M1 to M8 (refer to 3.2.1), confusion matrices implicitly provide class-level precision and recall insights.

Comparing training and test confusion matrices indicates that both models generalize well, with no evidence of severe overfitting. Random Forest shows slightly more stable behavior between training and test performance, the overall differences between the two models are small.

In addition, confusion matrix analysis reveals that classification performance varies across cancer types. One class (class 3) is consistently well classified, while another class (class 2) is significantly more difficult, with substantial misclassification into other classes. Importantly, both models exhibit nearly identical misclassification patterns, suggesting that the primary source of error lies in the data structure rather than the choice of model. Based on these observations, Random Forest was selected as the final model due to its slightly more stable generalization performance.

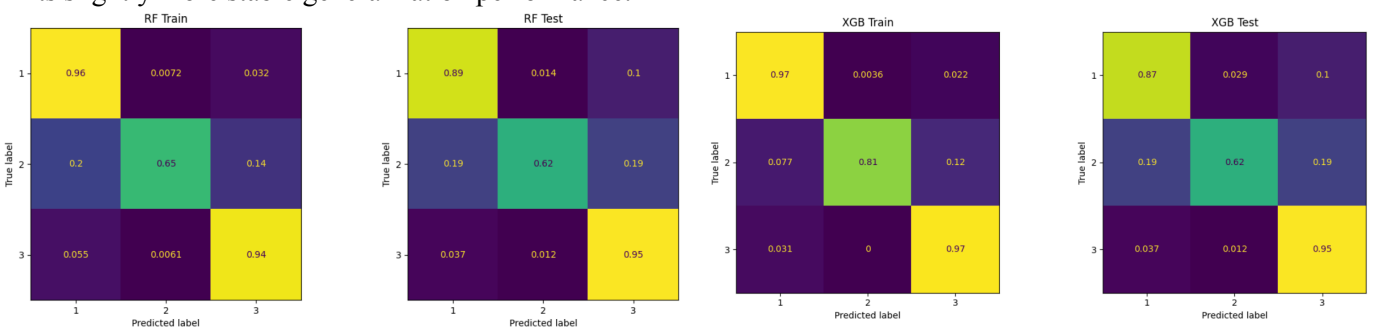

Figure 7. Confusion Matrix Comparison of Random Forest and XGBoost on Training and Test Sets

4.1.2 Impact of Label Noise on Random Forest (Mislabeled Analysis)

To further investigate the source of classification errors, we explored the impact of potential label noise in the dataset. Suspected mislabeled observations were identified based on model agreement and confidence: samples misclassified (pred ≠ y), with high prediction probability (> 0.85), and consistently predicted by both Random Forest and XGBoost were flagged as potentially mislabeled.

Figure 8 shows a substantial improvement in classification performance after excluding these observations. Refer to Figure 8, overall accuracy increased from approximately 0.8995 to 0.9432. The most significant improvement was observed in the previously challenging class (class 2), where accuracy increased from 0.65 to 0.81, accompanied by a notable reduction in misclassification rates. Improvements were also observed in class 1, while class 3 remained consistently well classified.

These findings indicate that label noise is a major factor limiting model performance. The results further suggest that classification errors are not primarily due to model limitations, but rather due to inconsistencies or inaccuracies in the dataset. This highlights the importance of data quality in high-dimensional biological classification tasks.

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RF Accuracy BEFORE drop: 0.899548532731377
RF Accuracy AFTER drop: 0.9431952662721893
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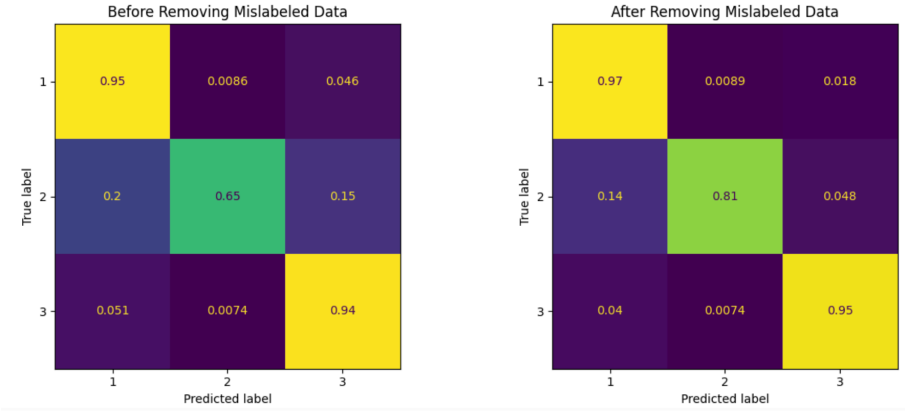

Figure 8. Impact of Removing Suspected Mislabeled Data on Random Forest Performance

4.2 Conclusion

The goal of this project is to develop a supervised classification model that accurately predicts three cancer types (GBM, LUSC, and OV) using high-dimensional gene expression data. Our results demonstrate that accurate cancer classification using gene expression data is achievable. Tree-based ensemble models, particularly Random Forest, achieve strong predictive performance, indicating that complex nonlinear relationships can be effectively captured in high-dimensional settings.

Meaningful gene features can be identified through LASSO-based feature selection, which reduces dimensionality while retaining important predictive signals. This contributes to improved model interpretability and helps address the challenges associated with high-dimensional data. In addition, feature importance analysis from the Random Forest model provides further insight into which genes contribute most to classification performance. The model also demonstrates good generalization performance, as evidenced by consistent results across training and test datasets. However, our findings highlight that model performance is not solely determined by model choice, but is strongly influenced by data quality. The significant improvement observed after excluding suspected mislabeled observations suggests that label noise is a major limitation in this task.

Overall, this project builds a model to ensure accurate cancer classification, while emphasizing the importance of feature selection, model stability, and data quality in high-dimensional classification modelling.

Feature Importance

Feature importance in the random forest classification model is computed using the feature's contribution to the mean decrease in the Gini index, which reflects the extent to which each feature reduces classification uncertainty across decision trees.

As shown in Figure 9, our model identifies genes including SIGLEC8, PTPN22, RXFP3, CALML3, and TP63 as the five most important genes in performing cancer type classification, which are supported by literature (Au et al., 2004; Ho et al., 2021; Huang et al., 2010; Liu et al., 2019; Trebo et al., 2021). Certain limitations exist when interpreting the outcome of feature importance in our model. Due to an imbalance in the number of samples per class, feature importance in our RF model may be biased towards genes that are important for classes with larger sample sizes. Moreover, feature importance only reflects correlation rather than causation (e.g. pathogenesis), which requires further biomedical investigations (more is discussed in the inspiration part of the next section).

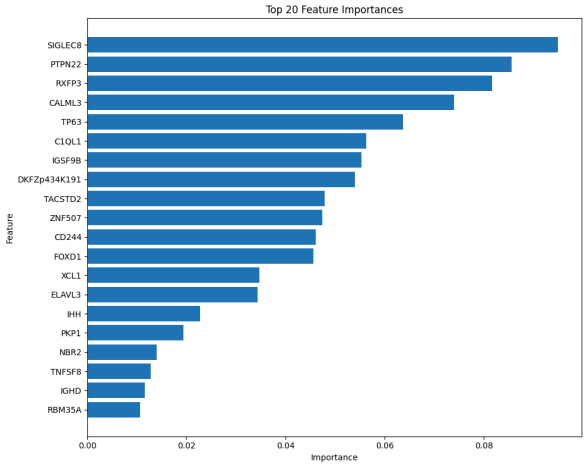

Figure 9. Top 20 Feature Importances in the Random Forest Model

5. Discussion

Limitations

Given this data set, the key challenge in this project is effective and reasonable feature selection. In this project, we resolve high-dimensionality by regularization with LASSO. Although LASSO is able to select features directly and provide interpretable models, it has several limitations in terms of high-dimensional biomedical data.

First of all, Genomic datasets contain highly correlated and linearly dependent variables, which will raise concerns about instability and inconsistency for existing variable selection methods like LASSO (Wang et al., 2019). LASSO tends to select one variable from a group of correlated predictors while ignoring others, which may lead to selection inconsistency and loss of biological information, as Wang suggests. In comparison to LASSO variable selection, dimension reduction methods can generate low-dimensional representations while preserving relevant information. For example, Li et al.(2004) propose a dimension reduction strategy, which combines principal components analysis (PCA) and sliced inverse regression (SLR), to identify linear combinations of genes. This combined method considers the variability in the gene expression levels and preserves the information, making this method more effective for predictive models in high-dimensional settings. However, dimension reduction often sacrifices model interpretability; therefore, future studies should consider the trade-off between feature selection and dimension reduction.

Moreover, to better fit the high-dimensional, small-sample setting, the selected minimum leaf size is larger than in classical random forests, which may reduce model flexibility and raise the underfitting risk.

Inspiration

When examining existing literature to support the biological relevance of our model, we observed a substantial imbalance in research attention across different genes. This phenomenon has previously been documented by Stoeger et. al. (2018) and explained as due to historical and structural biases in scientific investigation.

Gene \ Source	PubMed	Google Scholar
SIGLEC8	24	2,410
TP63	1624	16,400

Table 6. Publication counts for selected genes (query: "[gene] + cancer")

Genes that our random forest model identifies as more important are not necessarily the most studied in existing literature. For example, although our model relies more heavily on SIGLEC8 to make predictions, the number of publications about it is significantly lower than for TP63.

This observation suggests that our model may highlight potentially underexplored genes that could be relevant for cancer classification for further biomedical investigation. However, this observation should be interpreted with caution. Since the training dataset is imbalanced across cancer types, the model may place greater emphasis on features that are more informative for the majority class. Therefore, while our findings provide useful insights into potentially understudied genes, further validation using more balanced datasets may enhance the credibility when drawing biomedical conclusions.

6. References

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